

MECHANICAL ENGINEERING CANDIDATE

Joseph Dwyer

Systems builder. Automation · Robotics · Motion control · Motorsports electronics · Local AI

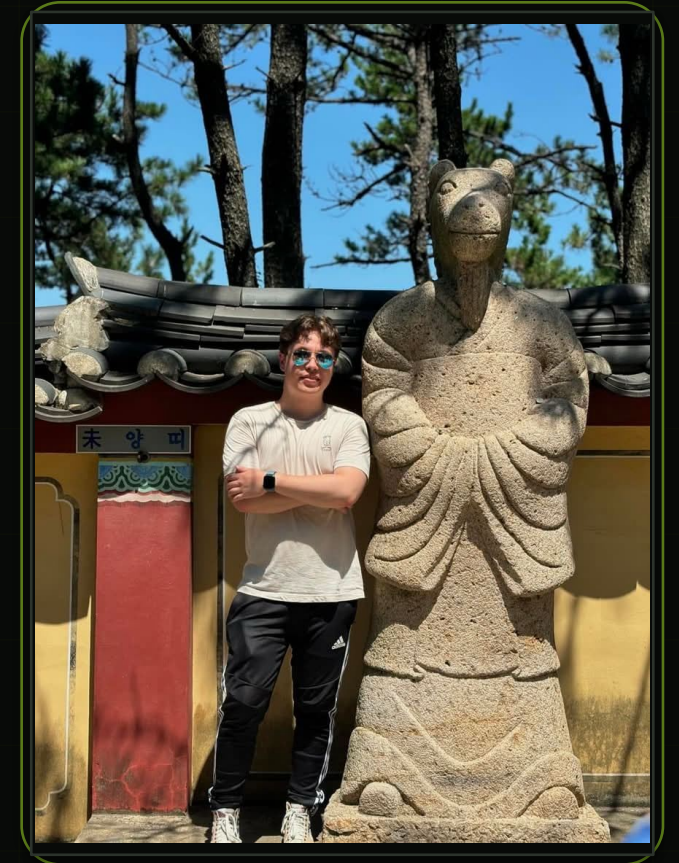
B70 AI Server

Small-business founder

3+ yrs hands-on

CSU Sacramento · ME

- 01 I build the whole stack — hardware, firmware, software, and the systems that run them — then document what I learn.
- 02 Work spans local AI infrastructure, CoreXY printer systems, ABB/Rexroth robotics, a senior capstone, and a student motorsports program I led from Treasurer to President.



ANTELOPE, CA

01 / B70

B70 AI Automation Server

Designed, built & run the whole stack — Intel Arc Pro B70

- › Built the server from scratch — enterprise components, custom 3D-printed frame, local-only inference over Tailscale.
- › Two-model runtime: Qwen3.8-27B heavy primary/manager + Qwen3.6-35B-A3B rapid worker, served via llama.cpp on SYCL.
- › Agent layer: Hermes Agent orchestrates real workflows — coding, monitoring, and the PFC Supervisor + Pulse dashboards.
- › 12+ benchmark runs + a full MTP-draft sweep to find the production profile.

PRODUCTION SERVING PROFILE

30.93 tok/s
QWEN3.8-27B DECODE ·
MTP-TUNED

192k
CONTEXT WINDOW

1.55 ×
MTP DRAFT LIFT OVER
NATIVE

F16
NATIVE VISION
(MMPROJ)



The B70 in its 3D-printed chassis — the server behind every system here

NATIVE BENCHMARKS — llama-bench

Full Library, Native Decode

model	params	pp512 t/s	tg128 t/s
Qwen3.5-0.8B Q8_0	0.75 B	14,262.06	202.79
Gemma-4-E2B Q4_K_XL	4.65 B	5,315.26	136.58
Gemma-4-E4B Q4_K_XL	7.52 B	3,165.46	88.48
Gemma-4-12B-it Q4_K_XL	11.91 B	1,511.79	49.04
Gemma-4-26B-A4B Q4_K_M	25.23 B	772.25	66.89
Qwen3.6-35B-A3B Q5_K_XL	34.66 B	724.97	75.63
Qwen3.6-35B-A3B Q5_K_S	35.51 B	724.90	72.77
Qwen3.6-35B-A3B Q4_K_M	34.66 B	706.68	77.52
Kanana-2-30B-A3B Q4_K_M	30.67 B	593.42	28.18
Muse-Glimmer-30B Q5_K_XL	27.85 B	569.33	22.00
Gemma-4-31B Q4_K_XL	30.70 B	427.72	22.62
Qwen3.8-27B Q6_K	27.32 B	485.92	19.92

HyperCLOVAX-SEED-Think-32B Q4_K_M — failed to load (2 s, no table emitted).

FINDINGS

- ▶ MoE wins decode: the three Qwen3.6-35B-A3B quants all sustain 72–78 t/s generation at ~24 GB — the library's fastest decode.
- ▶ Qwen3.8-27B Q6_K (the production model) sits at 19.92 t/s native — which is exactly why the MTP-draft work matters.
- ▶ Small models are embarrassingly fast on the B70: 0.8B does 202.8 t/s; even the 31B dense Gemma holds 22.6 t/s.

MTP SWEEP – multi-token prediction

Drafting: the 1.67× finding

model	mode	decode t/s	accept %	vs base
Qwen3.8-27B Q6_K	baseline	19.87	–	1.000×
Qwen3.8-27B Q6_K	mtp-n2	33.20	70.70	1.671×
Qwen3.8-27B Q6_K	mtp-n4	30.76	51.04	1.548×
Qwen3.8-27B Q6_K	mtp-n8	8.83	32.91	0.444×
Gemma-4-26B-A4B	mtp-n4	93.33	63.61	1.239×
Gemma-4-31B	mtp-n2	33.06	81.44	1.496×
Qwen3.6-35B-A3B	mtp-n2	65.31	81.00	0.847×

FINDINGS

- ▶ Best decode speedup in the sweep: Qwen3.8-27B mtp-n2 → 1.671× (19.87 → 33.20 t/s) at 70.70% accept.
- ▶ Best accept rate: Gemma-4-31B mtp-n2 → 81.44% (1.496× decode).
- ▶ Honest negative: the Qwen3.6-35B-A3B MTP build ran slower than its own baseline (0.847× at n2) — a fast MoE native decode leaves nothing for the draft to add.
- ▶ n8 degrades every model (0.44×–0.63×): deep speculative depth over-commits tokens that don't accept. Shallow n2 is the sweet spot.

PRODUCTION PROFILE — optimized MTP draft

19.92 → 30.93 tok/s

config	osl	prompt t/s	decode t/s	accept %
Qwen3.8-27B · n2 p0.00	1024	136.76	30.16	61.8
Qwen3.8-27B · n2 p0.15	1024	136.88	30.93	70.8
Qwen3.8-27B · n2 p0.25	1024	137.58	30.93	66.3
Qwen3.8-27B · n2 p0.35	4096	138.72	30.83	71.2
Qwen3.8-27B · n2 p0.50	1024	137.71	29.53	76.0
Qwen3.8-27B · n3 p0.70	4096	136.94	29.01	83.2
Qwen3.6-35B · baseline	1024	272.32	77.25	—
Qwen3.6-35B · mtp-n2	1024	263.26	65.29	81.0

Headline: the optimized draft lifts the production model from 19.92 t/s native to 30.93 t/s at n2 p0.15/p0.25 OSL1024 — ~1.55×, the best clean decode in the series.

Production model (Qwen3.8-27B) re-benchmarked at 192k context...

LIVE SERVING CONFIG

endpoint: local /v1 worker endpoint

model: Qwen3.8-27B Q6_K (+ optimized MTP draft in active profile)

vision: native · mmproj F16 (885 MB)

offload: --no-mmproj-offload (projector on CPU)

context: 192k

runtime: llama.cpp 10369 · SYCL · IntelLLVM 2026.1.1

network: local to B70 · Tailscale for remote

CONTROL PLANE

PFC Supervisor

control plane · Pulse command widget

A self-built control plane that supervises a Pi + PFP-host pair: telemetry, authenticated service operations, a model-agnostic concurrent AI runtime, a power-automation policy engine that puts idle machines to sleep, and a desktop/mobile 'Pulse' command widget for voice and one-tap fleet control — with a savings ledger that stays honest.

Target nodes: Raspberry Pi OS 13 + Fedora 44

API schema: 1.4.0

HTTP model: bounded 24-worker gate · PFC_HTTP_MAX_WORKERS

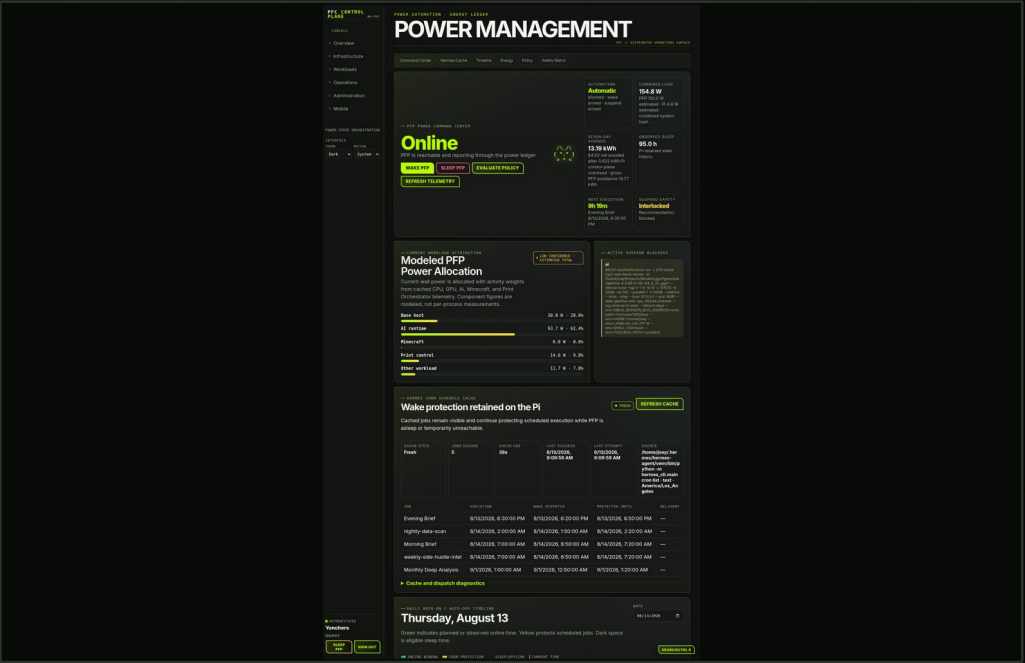
Backpressure: 503 SERVER_BUSY · Retry-After: 1 when saturated

GET dedupe: single-flight protection · 45s browser timeout

HDMI state: WAITING · ONLINE · STALE · FAULT (5 fails + 30s)

Concurrency: listen backlog 64→128 · 24-worker ceiling

Theming: light · dark · system · animated interface framework



154.8

W COMBINED LOAD

PFP 150.0 + Pi 4.8 W est.

18.3

W CORE ONE OUTLET

live plug telemetry

13.19

KWH 7-DAY AVOIDED

\$4.62 net · PFP 13.77

95.0

H OBSERVED SLEEP

Pi-retained history

68.23

KWH 30-DAY PROJ.

policy projection

FLEET OPS

Print Orchestrator

dashboard · 8090

A local fleet dashboard over mixed printers (PrusaLink + Moonraker) with unified jobs, server health, per-printer power, and camera-based part-presence failure detection.

Transports: PrusaLink (Core One) + Moonraker (Voron/Klipper)

Failure detect: camera part-presence scoring

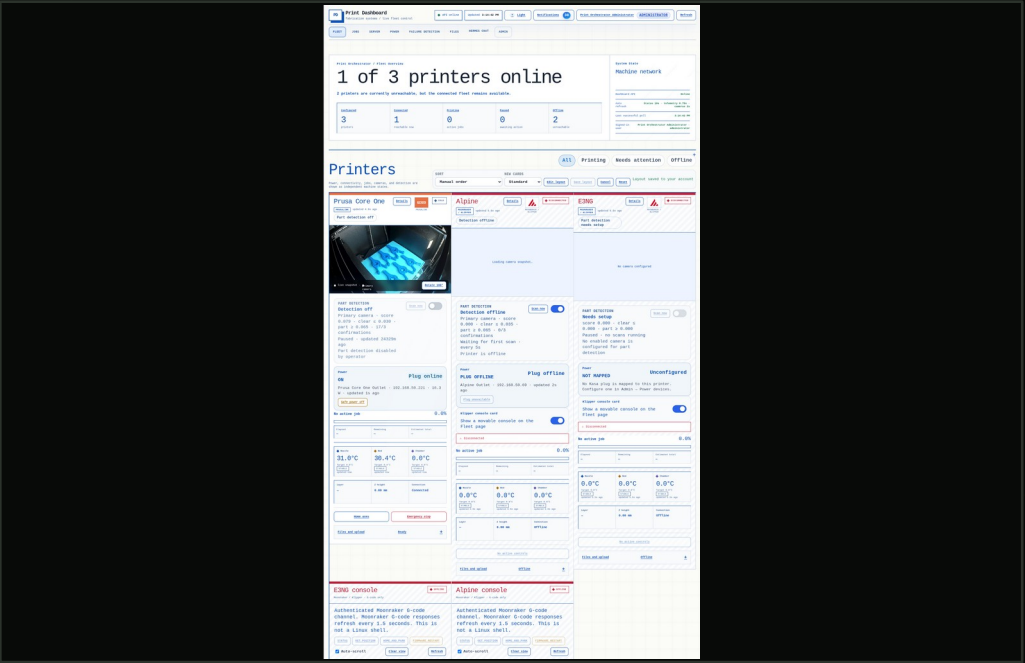
Confirmation: 17/3 confirmations on primary camera

Host: pfp-server · Tailscale reachable

Pages: Fleet · Jobs · Server · Power · Failure Detection · PrusaLink

Part gate: clear ≤ 0.030 · part ≥ 0.065

Power: per-outlet live · safe power off



- SCOPE**
- › multi-printer fleet with mixed PrusaLink / Moonraker backends
 - › unified job queue + progress + remaining-time estimation
 - › server health + worker accounting
 - › per-printer power with safe power-off
 - › camera part-presence failure detection with scoring thresholds
 - › Tailscale-exposed local dashboard (8090)

AGENT LAYER

Hermes / Odysseus

local-first agent workflows

A human-in-the-loop agent layer on the B70: email triage, human-reviewed draft generation, translation, meeting minutes, calendar support, research summaries, and supervised digital-manufacturing workflows — all local, all private.

Runtime: Hermes Agent · local models (Qwen/Gemma)

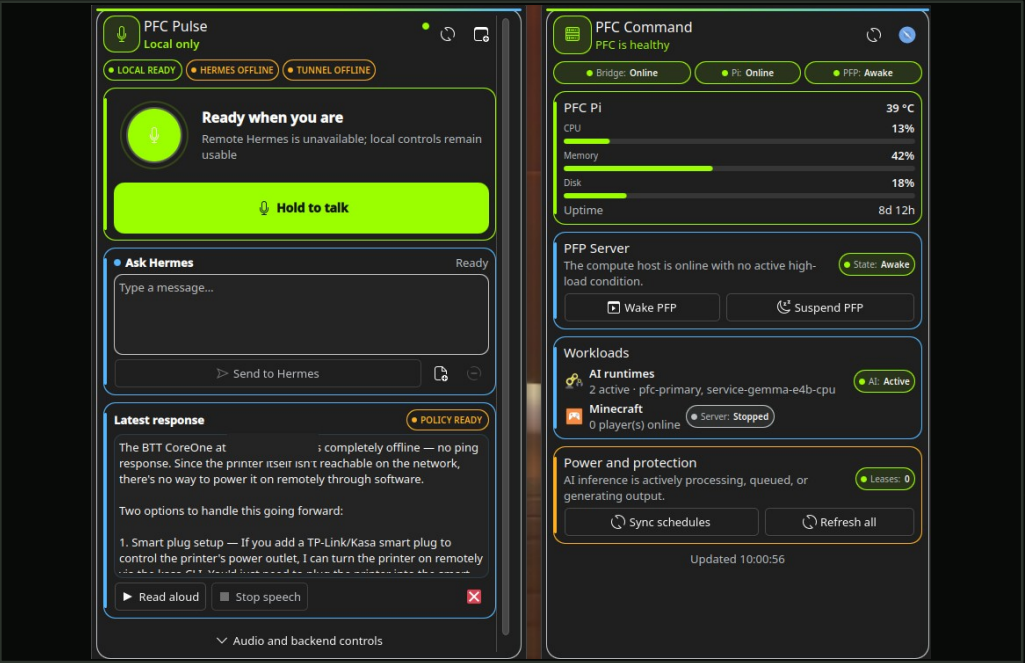
Network: local worker endpoint (/v1) · Tailscale

Workflows: email triage · minutes · translation · research

Model: Qwen3.8-27B Q6_K · native vision (mmpoj F16)

Review model: human-reviewed drafts · no silent auto-send

Print: slicer prep · job scheduling · status monitoring



SCOPE

- › email triage + human-reviewed draft generation
- › translation + meeting-minute generation
- › calendar support + technical research summaries
- › task review + documentation assistance
- › supervised slicer prep, printer job scheduling, status monitoring
- › local-first: engineering context stays on the tailnet

3D PRINT • KLIPPER

E3NG Custom CoreXY

Designed, printed & tuned the build

An RH3D E3NG build — an Ender 3 recycled into a CoreXY: reuses the original Ender 3's parts, then rebuilds it as a fast, enclosed, Klipper-driven CoreXY with a continuous eddy-current bed sensor, tuned and printed for engineering-grade parts.

Ender 3 → CoreXY (RH3D E3NG)

reuses the original Ender 3's parts

Klipper • enclosed build

eddy-current bed sensor (continuous)



DOCUMENTATION

The Ender 3 origin — The E3NG is a conversion, not a new machine: it takes an old, often-obsolete Ender 3 and recycles the bulk of its parts — frame hardware, leadscrews, the original electronics where it works — then...

The continuous bed sensor — Instead of a discrete click-probe that taps the bed at grid points, this build uses an eddy-current (eddy induction) sensor that measures bed height continuously as the carriage moves, producing a smooth...

3D PRINT • KLIPPER

Voron 2.4

Built, troubleshoot & dialed in

Built and troubleshoot a Voron 2.4 CoreXY from frame to first layer — stepper motion, toolhead and hotend integration, custom print profiles, and full-system diagnosis until the parts came off clean at speed.

Voron 2.4 CoreXY

Klipper • high-speed profiles

toolhead + hotend integration

custom profiles • diagnosis



DOCUMENTATION

The build — A from-scratch Voron 2.4 CoreXY: frame assembly, the internal electronics bay (PSU, drivers, fans, and the wiring that keeps the toolhead and bed in sync), stepper motion setup, and the toolhead...

The work — This is the build-and-fix layer: when something stops being straight, the loop is measure (calibration prints, the camera), tune (the Klipper profile), and verify — repeated until the part is right...

MECHANICAL DESIGN • TEAM

ME Senior Capstone — Espresso Machine

Team 1 • CAD, fab & assembly

ME190 senior capstone (Team 1 • Crema Crew): a hand-actuated espresso machine — CAD, worm-gear + linkage tamping mechanism, machined and 3D-printed parts, full assembly, and a final report + engineering poster.

ME190 • team capstone

hand-actuated espresso machine

CAD → fabrication → assembly

final report + ME190 poster



DOCUMENTATION

The machine — A hybrid hand-actuated espresso machine: a worm gear drives a linkage to give ~120:1 mechanical advantage, so a ~5 lb force on a hand lever produces the ~500 lb / 9 bar needed to pull a consistent double...

Scope — A full five-member team design cycle: requirement capture, CAD (SolidWorks/Fusion), part fabrication (machining + 3D printing for prototypes), assembly, and functional testing — closed with a final report...

ROBOTICS • CONTROLS

ME165 • ABB / Rexroth

Lab operation, motion & troubleshooting

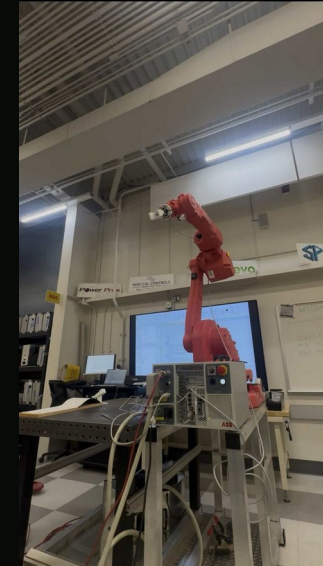
Hands-on academic robotics: ABB robot systems + Rexroth controller platforms — motion concepts, controller interaction, lab setup, troubleshooting, and automation documentation.

ABB robot cells

Rexroth controllers

lab setup + troubleshooting

workcell documentation



DOCUMENTATION

Scope — ME165 is the academic robotics core: ABB industrial robot cells and Rexroth controller platforms. The work is the full operator's loop — lab setup, controller interaction, motion programming, fault...

Why it's here — The direct foundation for the controls & automation line: reading a live controller, understanding motion, isolating a fault on a running cell, and writing the documentation so the workcell reproduces...

RACECAR ELECTRONICS · CDO

Metuned LLC

Chief Design Officer · 2023-2026

Co-founded a racecar data-acquisition startup (Chief Design Officer, 2023-2026): affordable performance telemetry, prototype iteration, low-cost manufacturability, and product documentation.

Chief Design Officer

data-acquisition / telemetry

prototype iteration

low-cost manufacturing



DOCUMENTATION

The product — Metuned made affordable performance data acquisition for racecars: a compact DAQ box that captures telemetry from a race vehicle's harness, a dashboard for live and post-run review, and cage-mount...

The constraint set — The interesting engineering is cost: every part choice is made against 'can a customer afford this?' — low-cost manufacturability is a design requirement, not a later optimization. Prototype iteration...

TEAM LEADERSHIP · BUILD

Motorsports Club · 24h Lemons

Club officer · led the build

President/VP/Treasurer of the Sierra College Motorsports Club — led a 24 Hours of Lemons Mini Cooper build: race prep, sponsorship/outreach, funding (including a 3D-printed keychain fundraiser), welding, and real-time race support.

24 Hours of Lemons

Mini Cooper build

sponsorship + funding

Cars & Coffee keychain fundraiser

real-time race support

24h
LEMONS RACE

17
WRAP-UP
REPORT

3
OFFICER
ROLES

\$500
CAR BUDGET



DOCUMENTATION

The role — As President/VP/Treasurer of the Sierra College Motorsports Club I ran both sides of a 24 Hours of Lemons Mini Cooper build: the engineering side (race prep, roll-cage welding, engine bay, real-time race...

Cars & Coffee — the keychain fundraiser — One funding channel the club ran was 3D-printed car keychains for donation at Cars & Coffee events: I designed the pendant and ran the print jobs (the CR-10 Max V3 handled the runs), then sold them...

THE BUILD • 24h LEMONS

From Garage to Sonoma

A student Mini Cooper built for 24 Hours of Lemons — welded cage, 3D-printed chassis, race prep, and a crew that ran it all day at Sonoma.



Replacement pick-and-pull motor



Welding the roll cage



Club go-kart — club days recruitment



VR sim rig — bucket seat & harness



On track



Pit support



Student project — exo-set Miata dash



Fabricating the transmission mount



Full crew at Sonoma

TEST & MEASUREMENT

UEI Lab • Energy Testing

Lab assistant • measurement & docs

Lab assistant supporting CEC appliance energy-efficiency audits, portable-spa testing, and residential fan airflow/CFM testing aligned to HVI 916 — controlled environments, repeatability, documentation.

CEC energy audits

HVI 916 fan/CFM

test-environment build

data + documentation



DOCUMENTATION

The work — At University Enterprises Inc. (Sacramento State) I support CEC appliance energy-efficiency audits, portable-spa testing, and residential fan airflow/CFM testing aligned to HVI 916. The through-skill...

Connection to the PFC — This lab work is the mechanical origin of the PFC power policy engine: measuring real appliance draw, understanding what 'avoided energy' actually means, and keeping a ledger honest. The 13.19 kWh / 7-day...

EXPERIENCE

The Timeline

- May 2026 - Present

■

Lab Assistant

· University Enterprises Inc. / Sacramento State

CEC appliance energy-efficiency audits, portable-spa testing, and residential fan airflow/CFM testing (HVI 916). Controlled test environments, instrumentation, data collection, repeatability, and technical documentation.
- Jul 2023 - Jan 2026

■

Chief Design Officer

· Metuned LLC

Co-founded a racecar electronics startup for affordable data-acquisition/performance solutions. Prototype planning, testing, mechanical/electrical design direction, product documentation, low-cost manufacturability.
- Apr 2023 - Dec 2023

■

Makerspace Technical Assistant

· Sierra College

Trained users on 3D printers, CNC routers, laser cutters, and shop equipment; maintained, organized, and troubleshot makerspace equipment while enforcing safety standards.
- Jul 2021 - Sep 2023

■

Inventory / Packaging / Deburr Technician

· Vanquish Products / ASB Products

Managed inventory flow for production parts, coordinated anodizing/laser-etch processing, and supported shipping, packaging, deburr, and quality during high-demand periods.
- May 2023 - Dec 2024

■

President / VP / Treasurer

· Sierra College Motorsports Club

Led race-prep for a 24 Hours of Lemons Mini Cooper build; coordinated student tasks, managed sponsorship/outreach, and improved team funding through events and industry partnerships.

EDUCATION

California State University, Sacramento

B.S. Mechanical Engineering

Dec 2024 - Present · exp. Fall 2026 · 3.358

Sierra College

A.S. Natural Science + A.S.-T Engineering

Jul 2019 - Dec 2024 · 2.73 · 30-40 hrs/wk

WORK

Lab Assistant · UEI / Sacramento State

2026 - Present · CEC energy audits · HVI 916 fan/CFM testing · documentation

Chief Design Officer · Metuned LLC

2023 - 2026 · Co-founded racecar DAQ startup · product + ME design direction

Makerspace Tech Assistant · Sierra College

2023 · Trained users on 3D printing, CNC, laser · equipment + safety

Packaging / Deburr Tech · Vanquish Products

2021 - 2023 · Production inventory flow · anodizing + etch coordination · QA

SKILLS

▶ CONTROLS & AUTOMATION

ABB/Rexroth robotics, motion control, sensors, low-voltage troubleshooting, commissioning, fault isolation, root-cause analysis

▶ DIGITAL MFG / OT

PLC/HMI/SCADA fundamentals, industrial controls, OT/networking, Tailscale, Linux, local-AI workflow automation, human-reviewed agents

▶ CAD/CAM & FAB

Fusion 360, SolidWorks, AutoCAD, CNC routing, Haas CNC, milling, lathe, laser cutting, deburr, rapid prototyping, 3D printing

▶ PROGRAMMING & DATA

Python, C, Java, MATLAB, Klipper, llama.cpp, Qwen/Gemma local models, test data collection, engineering documentation

▶ COMMUNICATION

technical training, operator/user support, vendor coordination, shop safety, procedure writing, project planning, sponsorship/outreach

SYSTEMS ARCHITECTURE

The Systems Map

One B70 server runs the whole stack — the agent, the control planes, the tailnet — and manages a printer + power fleet. Everything local, everything on one tailnet.

ON THE B70

pfp-server

Hermes / Odysseus

agent workflows · human-reviewed · local inference

Print Orchestrator

fleet ops · job queue · failure detection · 8090

Tailscale

tailnet · pfp-server ↔ fedora client

▼ runs on

THE CORE

one box

B70 AI Server

local inference core

Intel Arc Pro B70

Fedora 44 · llama.cpp 10369

Qwen3.8-27B · Q6_K

3D-printed chassis

100% local · private

▼ manages — PFC + Orchestrator control planes

FLEET HW

managed

Raspberry Pi + PFC

energy ledger · power policy · voice · WoL telemetry

Prusa Core One

Prusa CoreXY · PrusaLink · 18.3 W

Voron / E3NG

Klipper · CoreXY

Kasa Smart Plugs

PFC power policy · live W

software runs on the B70 · fleet hardware is managed via control planes, not direct links · 100% local

Let's build the next system.

Local-first · hardware + firmware + software · documented end-to-end

yonchers.github.io

jdwyer@csus.edu

joeyoungdwyer@gmail.com

916-755-9100

linkedin.com/in/joseph-dwyer-10a4131ba